M3 | Water-hydraulic motor

Compact, water-lubricated axial-piston technology for a defined operating duty.

M3 converts water pressure into shaft torque and rotary motion. Size the starting load separately from the continuous duty, and match flow to speed and displacement.

Published parameter	M3 range rating
Maximum displacement (cm³/rev)	3.0
Minimum displacement (cm³/rev)	Not listed
Maximum speed (rpm)	4000
Minimum speed (rpm)	500
Maximum continuous power (kW)	2.7
Maximum input flow (L/min)	12
Maximum continuous pressure (bar)	160
Mass (kg)	1.5
Maximum temperature (°C)	50
Minimum temperature (°C)	2

Nominal ordering displacements: 3 cm³/rev. Nominal options and range-table limits are different source entries; use the selected configuration.

Design & environmental advantages

Multiple pistons smooth the delivery of power. Water lubricates the working surfaces, removing hydraulic oil from this circuit. Standard stainless construction and selectable internal materials support fluid-specific selection.

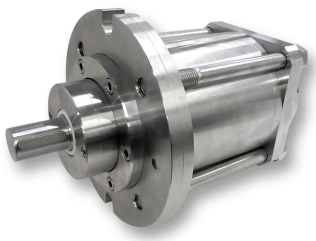
Efficiency & selection

Match component size to useful work. Read the supplied efficiency curve at pressure and speed. Avoid sustained relief flow and excessive throttling; these convert useful hydraulic power into heat.

Starting, overrun and procurement

Manufacturer starting-load guidance is 60% of dynamic torque. M3/M6 need at least 25 bar unloaded; larger models are stated to start above 12 bar. Specify load inertia, direction, case drain and overrun protection. Support radial and axial loads externally. Variable versions: M60 and M180.

Maxima are separate catalogue limits. Obtain the exact configuration drawing, duty-point selection, seal/service schedule, material documentation and applicable approval evidence with the quotation.



Family photograph; configuration varies.

Connect the selected design

Two-port motors are unidirectional; the return line maximum is 2.5 bar. Three-port models provide a separate case drain; return pressure must exceed leakage-port pressure, which is limited to 2.5 bar.

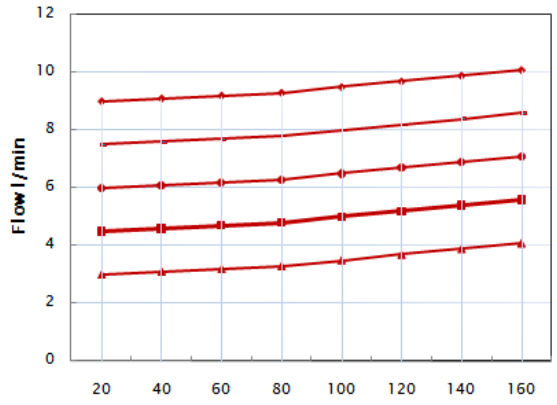
Reference conversions: 160 bar = 2,321 psi; 1 L/min = 0.264172 US gpm; 1 kg = 2.20462 lb.

Axial-Piston-Water-Motor-Full-Range-V.01.25.pdf, pp. 2-4. Website product context checked 17 Oct 2026.

M3 | Performance curves

Read displacement, speed and pressure together. Published curve data are preserved.

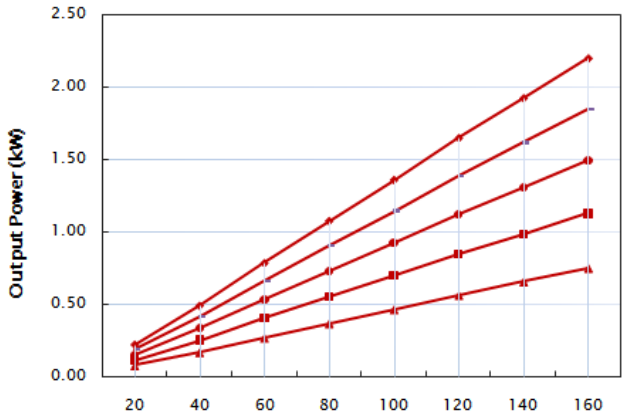
Input flow (L/min)



Pressure (bar)

3cc 1000rpm 3cc 1500rpm 3cc 2000rpm 3cc 2500rpm 3cc 3000rpm

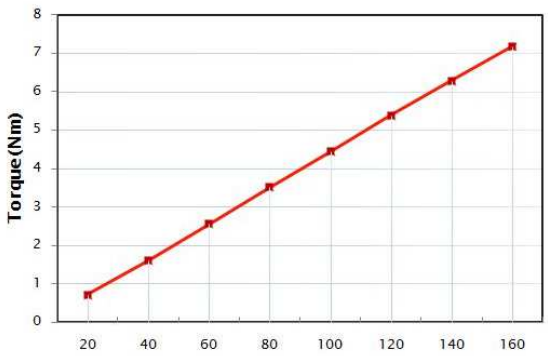
Power (kW): source label conflict



Pressure (bar)

3cc 1000rpm 3cc 1500rpm 3cc 2000rpm 3cc 2500rpm 3cc 3000rpm

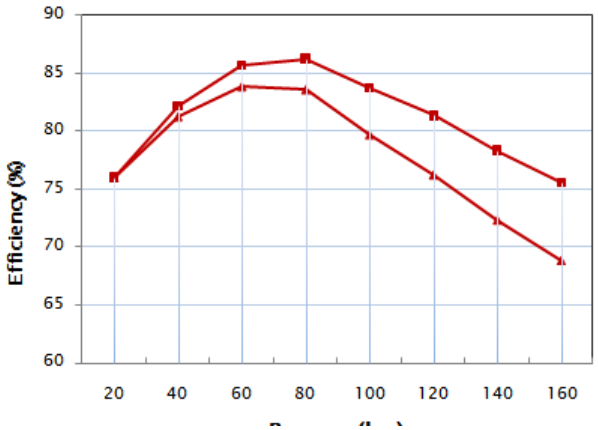
Output torque (N·m)



Pressure (bar)

3cc 1500rpm

Overall efficiency (%)



Pressure (bar)

3cc 1000rpm 3cc 1500rpm

Apply the curves to the duty

The source power chart heading says output power, while its axis says input power. Values are retained and the conflict is marked; obtain clarification before sizing from that chart.

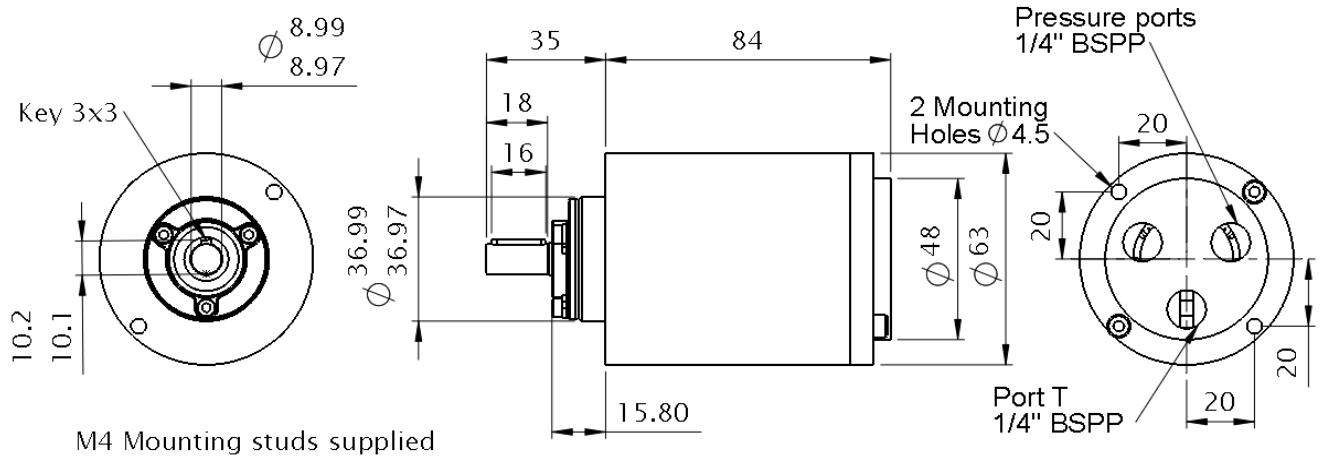
1 L/min = 0.264172 US gpm; 1 bar = 14.5038 psi; 1 kW = 1.34102 hp; 1 N·m = 0.737562 lbf·ft.

Axial-Piston-Water-Motor-Full-Range-V.01.25.pdf, p. 6.

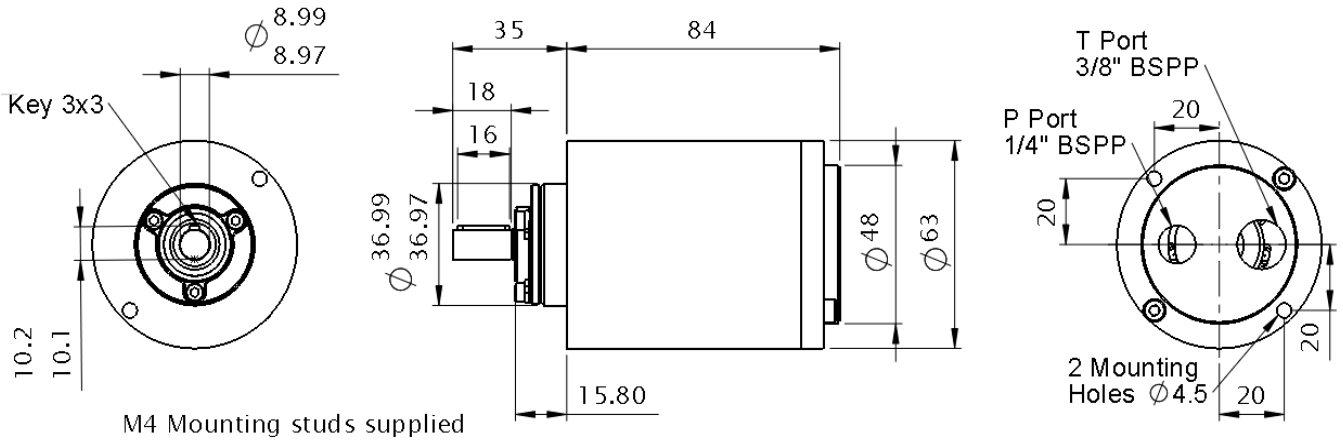
M3 | Dimensions & connections

Manufacturer technical reference | dimensions govern; reproduction is not to scale.

MB160-3.0W



MA160-3.0W & MC160-3.0W



Port threads and flange classes are connection designations, not converted diameters or operating-pressure ratings. Select the exact configuration shown on the approved order drawing.

Axial-Piston-Water-Motor-Full-Range-V.01.25.pdf, p. 5.